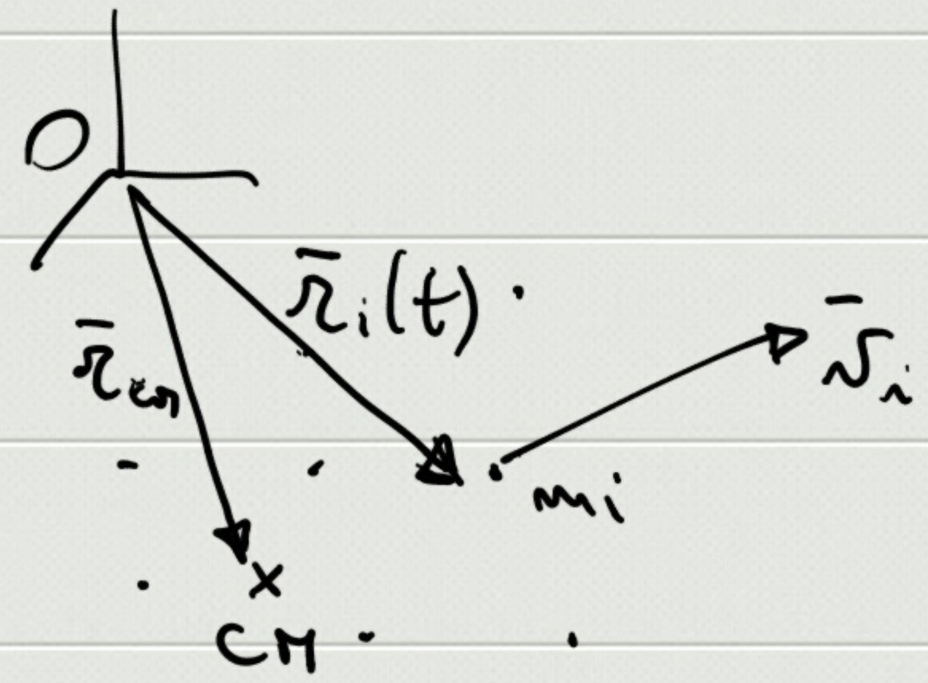
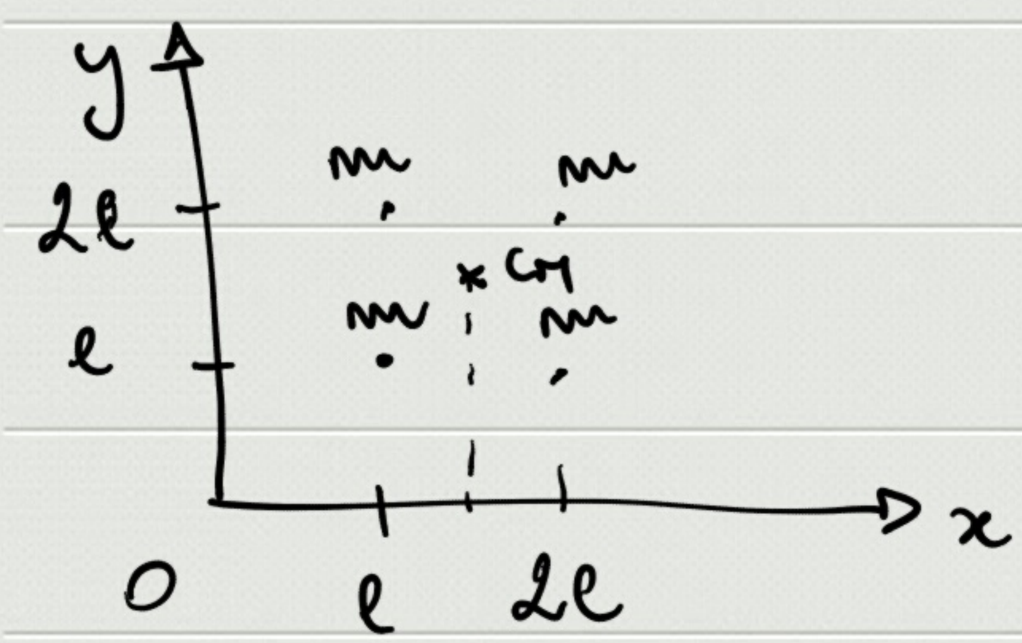
Centro di massa



$$\bar{r}_{CM} = \frac{m_1\bar{r}_1 + m_2\bar{r}_2 + \dots m_i\bar{r}_i + \dots + m_N\bar{r}_N}{m_1 + m_2 + \dots + m_i + \dots + m_N} =$$

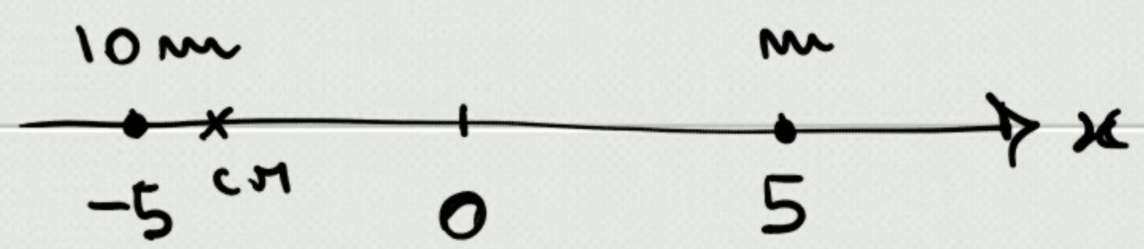
$$= \frac{\sum_i m_i \bar{r}_i}{\sum_i m_i} = \frac{\sum_i m_i \bar{r}_i}{m_{TOT}} \Rightarrow \begin{cases} x_{CM} = \dots \\ y_{CM} = \dots \\ z_{CM} = \dots \end{cases}$$



$$x_{CM} = \frac{ml + m2l + m2l + ml}{4m} =$$

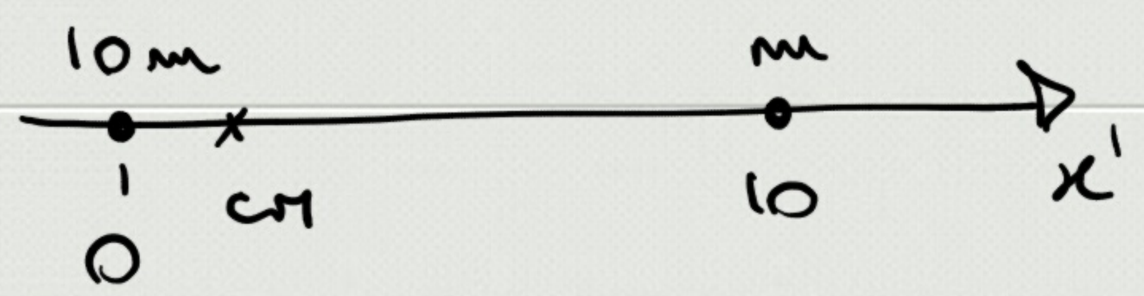
$$= \frac{6\cancel{m}l}{4\cancel{m}} = \frac{3}{2}l$$

$$y_{CM} = \frac{3}{2}l$$



$$x_{CM} = \frac{10m\cdot(-5) + m\cdot 5}{11m} =$$

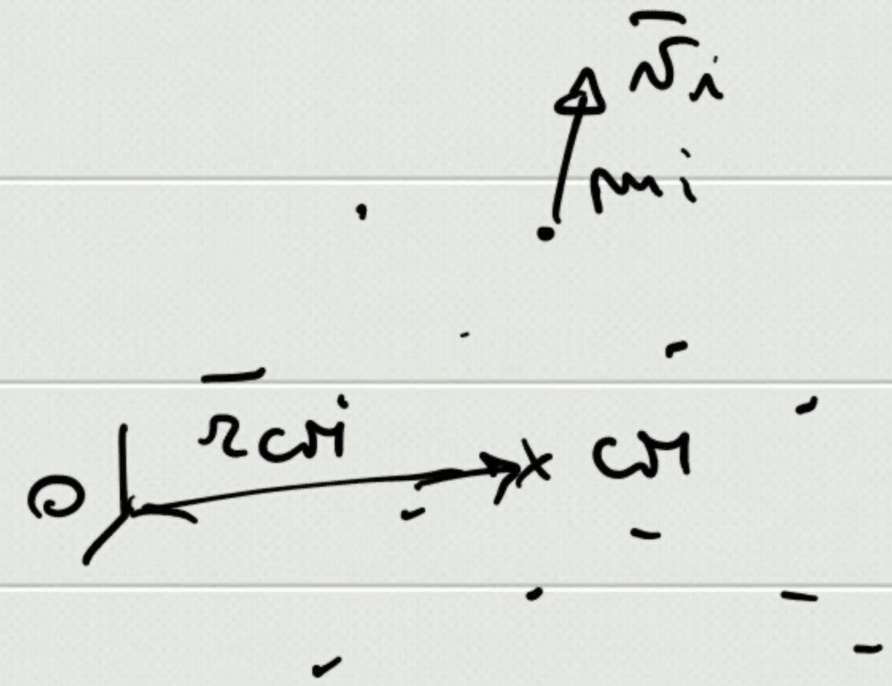
$$= -\frac{45}{11}$$



$$x'_{CM} = \frac{10m\cdot\emptyset + m\cdot 10}{11m} =$$

$$= \frac{10}{11}$$

$$\boxed{x = x' - 5}$$



$$\bar{r}_{CM} = \frac{\sum_i m_i \bar{r}_i}{\sum_i m_i} = \bar{r}_{CM}(t)$$

$$\bar{v}_{CM} = \frac{d\bar{r}_{CM}}{dt} = \frac{d}{dt}\left(\frac{\sum_i m_i \bar{r}_i}{\sum_i m_i}\right) = \frac{\sum_i m_i \frac{d\bar{r}_i}{dt}}{\sum_i m_i} = \frac{\sum_i m_i \bar{v}_i}{\sum_i m_i}$$

$$\Rightarrow \bar{v}_{CM} = \frac{\sum_i \bar{p}_i}{\sum_i m_i} = \frac{\bar{P}}{m_{TOT}} \Rightarrow \boxed{\bar{P} = m_{TOT}\,\bar{v}_{CM}}$$

$$\bar{a}_{CM} = \frac{d\bar{v}_{CM}}{dt} = \frac{d}{dt}\left(\frac{\sum_i m_i \bar{v}_i}{\sum_i m_i}\right) = \frac{\sum_i m_i \frac{d\bar{v}_i}{dt}}{\sum_i m_i} = \frac{\sum_i m_i \bar{a}_i}{\sum_i m_i}$$

$$\bar{a}_{CM} = \frac{\sum_i \bar{F}_i}{\sum_i m_i} = \frac{\sum_i (\bar{F}_i^{\,I} + \bar{F}_i^{\,E})}{\sum_i m_i} = \frac{\overset{0}{\cancel{\bar{R}^I}} + \bar{R}^E}{\sum_i m_i} = \frac{\bar{R}^E}{m_{TOT}}$$

$$\Rightarrow \boxed{\bar{R}^E = m_{TOT}\,\bar{a}_{CM}}$$

Teorema del moto del centro di massa

$$\bar{v}_{CM} = \frac{\bar{P}}{m_{TOT}}$$

$$\Rightarrow \bar{a}_{CM} = \frac{d\bar{v}_{CM}}{dt} = \frac{d}{dt}\left(\frac{\bar{P}}{m_{TOT}}\right) = \frac{1}{m_{TOT}}\frac{d\bar{P}}{dt}$$

$$\Rightarrow \boxed{\bar{R}^E = \frac{d\bar{P}}{dt}}$$

$\bar{R}^E = m_{TOT}\,\bar{a}_{CM}$ $\qquad$ $\bar{a}_{CM} = \dfrac{d\bar{v}_{CM}}{dt}$

$$\boxed{\bar{R}^E = 0} \Rightarrow \bar{a}_{CM} = 0 \Rightarrow \boxed{\bar{v}_{CM} = \text{cost}}$$

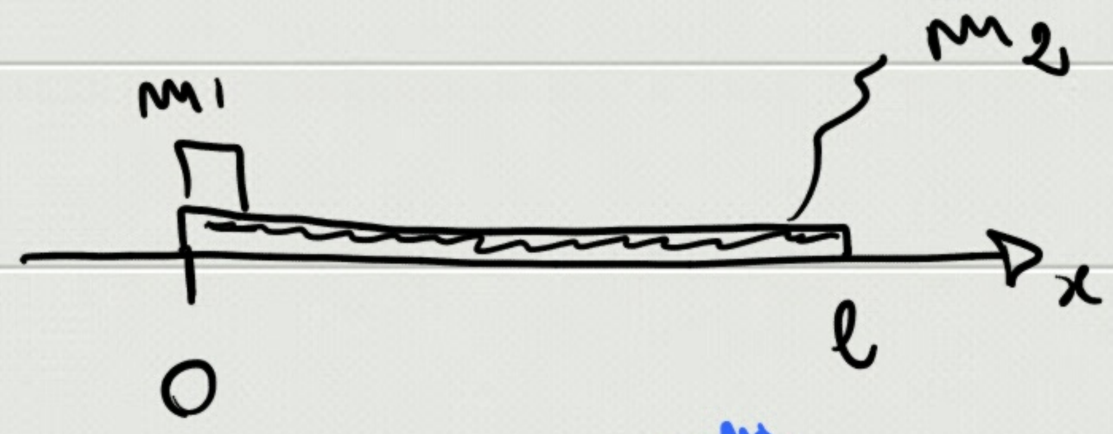
$$\Rightarrow \boxed{\bar{P} = \text{cost}}$$

Statica $\Rightarrow \boxed{\bar{v}_{CM} = 0}$

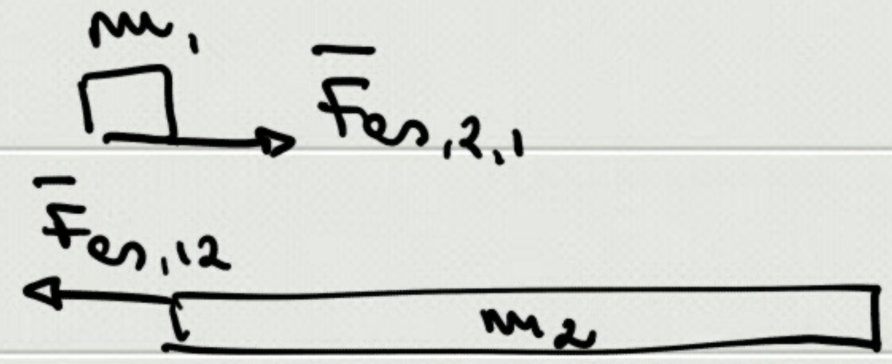
$$\bar{v}_{CM} = \frac{d\bar{r}_{CM}}{dt} = 0 \Rightarrow \boxed{\bar{r}_{CM} = \text{cost}}$$

$\bar{R}^E \neq 0$ ma $R^E_{\substack{x\\y\\z}} = 0 \Rightarrow P_{\substack{x\\y\\z}} = \text{cost}$

Sistema isolato



$$\boxed{R^E = 0} \quad (\bar{P}, \bar{N})$$



$$\Rightarrow \boxed{\bar{P} = \text{cost}} \underset{=0}{=} m_1 \bar{v}_1 + m_2 \bar{v}_2 \Big\} \Rightarrow \boxed{\bar{v}_2 = -\frac{m_1}{m_2} \bar{v}_1}$$

$(v_{10} = v_{20} = 0)$

$$\bar{P} = 0 \overset{!}{=} m_{TOT} \bar{v}_{CM} \Big\} \Rightarrow \bar{v}_{CM} = 0 \overset{!}{=} \frac{d\bar{r}_{CM}}{dt} \Big\} \Rightarrow \boxed{\bar{r}_{CM} = \text{cost}}$$

$$\Rightarrow \boxed{x_{CM} = \text{cost}}$$

$$x_{CM}^{i} = \frac{m_1 \cdot \emptyset + m_2 \frac{l}{2}}{m_1 + m_2}$$

$$x_{CM}^{f} = \frac{m_1 \cdot d + m_2 \left[\frac{l}{2} - (l-d)\right]}{m_1 + m_2} = \frac{m_1 d + m_2 \left(d - \frac{l}{2}\right)}{m_1 + m_2}$$

$$x_{CM}^{i} = x_{CM}^{f} \Rightarrow m_2 \frac{l}{2} = m_1 d + m_2 d - m_2 \frac{l}{2} \Rightarrow$$

$$\boxed{d = \frac{m_2}{m_1 + m_2} l}$$